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Chapter 1

Viscoelastic Fluid Mechanics and the Four-Roll Mill

In this chapter, the continuum mechanics framework governing the flow of incompressible viscoelastic fluids is established. The primary objective is to formulate the governing equations and constitutive models that describe the physical system within the four-roll mill geometry. We begin from the universal conservation principles of mass and linear momentum, introducing the physical and kinematic simplifications characteristic of polymer solutions under creeping and low-inertia flow conditions. The Oldroyd-B constitutive model is formulated by examining the Upper-Convected Time Derivative (UCTD), detailing its extensional kinematics and the origins of the High-Weissenberg Number Problem (HWNP). Non-linear closures, specifically the Giesekus and Phan-Thien–Tanner (PTT) models, are presented as distinct rheological extensions that account for anisotropic hydrodynamic drag and stress-dependent macromolecular entanglement breakdown while preserving the underlying linear structure. A viscous stress scaling procedure is introduced to define the governing dimensionless groups (Re_{scale} , Re_{phys} , Wi , β). Finally, the four-roll mill benchmark is formalized through its local kinematics, flow classification, geometry, boundary conditions, and the definition of

the forward and inverse problems.

1.1 General Conservation Laws

From a continuum mechanics perspective, the spatial and temporal motion of a fluid continuum is governed by the universal conservation laws of mass and linear momentum [1].

1.1.1 Principle of Mass Conservation

The conservation of mass for a fluid with local density $\rho(\mathbf{x}, t)$ and velocity vector $\mathbf{u}(\mathbf{x}, t)$ is expressed through the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \quad (1.1)$$

Equation (1.1) dictates that the instantaneous rate of mass accumulation within any fixed control volume must be strictly balanced by the net convective mass flux through its boundary surface.

1.1.2 Conservation of Linear Momentum

Applying Newton's second law to an infinitesimal fluid parcel yields the momentum equation in Eulerian coordinates:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}, \quad (1.2)$$

where $\mathbf{g}$ denotes the acceleration due to external body forces, and $\boldsymbol{\sigma}$ is the symmetric second-order stress tensor. The left-hand side represents the substantial derivative $D\mathbf{u}/Dt$, which combines the local unsteady rate of change $\partial \mathbf{u}/\partial t$ with the convective acceleration term $\mathbf{u} \cdot \nabla \mathbf{u}$ arising from spatial velocity gradients.

To isolate the isotropic pressure field from internal stresses generated by deformation, the total stress tensor is partitioned as:

$$\boldsymbol{\sigma} = -p\mathbf{I} + \mathbf{T}, \quad (1.3)$$

where p is the isotropic pressure field, $\mathbf{I}$ is the second-order identity tensor, and $\mathbf{T}$ is the extra-stress tensor. The extra-stress tensor encapsulates all non-hydrostatic stresses associated with the dissipative and elastic response of the fluid microstructure. Substituting Equation (1.3) into Equation (1.2) yields:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g}. \quad (1.4)$$

1.2 Physical Assumptions and Governing System Reduction

The full three-dimensional, unsteady equations of motion can be reduced by introducing scale-appropriate physical assumptions for polymer solutions within the four-roll mill geometry:

1. **Continuous medium and state of rest:** The fluid is modeled as a continuous medium that is locally isotropic in its thermodynamic state of rest. Under kinematic deformation, flow-induced macromolecular orientation and stretching break this resting isotropy, generating stress anisotropies captured by the extra-stress tensor.
2. **Incompressibility and stream function formulation:** Under the moderate operating pressures and low flow velocities characteristic of liquid polymer systems, density variations are negligible ($\rho = \text{const}$). The mass

conservation equation (1.1) simplifies to the divergence-free condition:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \quad (1.5)$$

In two-dimensional planar motion, this incompressibility constraint can be satisfied identically by introducing a scalar stream function $\psi(x, y)$ defined such that:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (1.6)$$

By Schwarz's theorem on mixed partial derivatives, $\nabla \cdot \mathbf{u} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} \equiv 0$ holds identically across the entire domain.

3. **Isothermal operating conditions:** Under the operating conditions considered in this study, internal temperature variations due to viscous dissipation remain negligible. The energy conservation equation decouples from momentum, allowing transport properties (viscosities, relaxation time) to be treated as constants.
4. **Absence of body forces:** The four-roll mill geometry is analyzed in a horizontal cross-sectional plane (x, y) . Gravitational acceleration acts along the vertical cylinder axis (z) , perpendicular to the modeled planar flow field. Consequently, body force contributions vanish in the modeled plane, and we set $\mathbf{g} = \mathbf{0}$.
5. **Steady-state flow:** Once constant operating conditions are established and the flow reaches a steady regime, all local temporal derivatives vanish identically ($\partial(\cdot)/\partial t = \mathbf{0}$).
6. **Two-dimensional planar kinematics:** Given the high aspect ratio of the cylindrical rollers along the axial dimension (z -axis), edge effects in the transverse mid-plane of the cavity are negligible. The velocity field is

therefore strictly planar:

$$\mathbf{u} = (u(x, y), v(x, y), 0). \quad (1.7)$$

Under these physical simplifications, the momentum equation reduces to:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \nabla \cdot \mathbf{T}. \quad (1.8)$$

1.3 Constitutive Modeling of Viscoelastic Fluids

In two dimensions, the momentum balance (1.8) contains six unknown scalar fields $(u, v, p, T_{xx}, T_{xy}, T_{yy})$, whereas mass and momentum conservation provide only three equations. A constitutive relationship for the extra-stress tensor $\mathbf{T}$ is required to close the system.

For dilute and semi-dilute polymer solutions, the extra-stress tensor $\mathbf{T}$ is partitioned into a purely viscous Newtonian solvent contribution and an elastic polymeric contribution $\boldsymbol{\tau}$ [1, 2]:

$$\mathbf{T} = 2\mu_s \mathbf{D} + \boldsymbol{\tau}, \quad (1.9)$$

where μ_s represents the dynamic viscosity of the Newtonian solvent, and $\mathbf{D}$ denotes the symmetric rate-of-strain tensor:

$$\mathbf{D} = \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]. \quad (1.10)$$

This decomposition isolates the instantaneous dissipative response of the solvent ($2\mu_s \mathbf{D}$) from the elastic stress $\boldsymbol{\tau}$ generated by macromolecular stretching and

orientation. Substituting Equation (1.9) into Equation (1.8) and exploiting incompressibility ($\nabla \cdot (2\mu_s \mathbf{D}) = \mu_s \nabla^2 \mathbf{u}$), the steady momentum equation becomes:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau}. \quad (1.11)$$

1.3.1 Upper-Convected Time Derivative

Polymeric liquids exhibit memory: macromolecular chains deform, orient, and relax over characteristic finite timescales. When a fluid element moves through a complex flow, its internal stress state is simultaneously advected along streamlines, rotated by local vorticity, and stretched by velocity gradients.

To satisfy the principle of material frame-indifference, constitutive equations must employ a frame-invariant tensor rate. For flexible polymer chains undergoing affine deformation with the surrounding continuum, the appropriate objective derivative is the *Upper-Convected Time Derivative* (UCTD), denoted by $\overset{\nabla}{\boldsymbol{\tau}}$. With the velocity gradient defined as $(\nabla \mathbf{u})_{ij} = \partial u_i / \partial x_j$, the upper-convected derivative of the symmetric extra-stress tensor $\boldsymbol{\tau}$ is:

$$\overset{\nabla}{\boldsymbol{\tau}} \equiv \frac{\partial \boldsymbol{\tau}}{\partial t} + \mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T. \quad (1.12)$$

Under steady conditions, $\partial \boldsymbol{\tau} / \partial t = \mathbf{0}$. The term $\mathbf{u} \cdot \nabla \boldsymbol{\tau}$ accounts for convective spatial transport of stress along flow streamlines, while the terms $-(\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T$ describe the stretching and rotation of macromolecular conformations induced by local velocity gradients.

1.3.2 The Oldroyd-B Model and Extensional Behavior

The classical Oldroyd-B constitutive model [4] represents macromolecules as dilute, non-interacting Hookean dumbbells suspended in a Newtonian solvent [1].

In dilute solution theory, the polymeric viscosity contribution μ_p is directly coupled to the relaxation time λ through kinetic theory:

$$\mu_p = nk_B T \lambda, \quad (1.13)$$

where n is the number density of polymer chains, k_B is the Boltzmann constant, and T is absolute temperature. Under steady conditions, the polymeric extra-stress tensor $\boldsymbol{\tau}$ satisfies:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}. \quad (1.14)$$

The total zero-shear viscosity of the fluid is $\mu_0 = \mu_s + \mu_p$.

While Oldroyd-B predicts a constant shear viscosity $\eta(\dot{\gamma}) = \mu_0$ (characteristic of ideal elastic fluids), it exhibits an intrinsic limitation in extensional kinematics. In a steady planar extensional flow ($u = Ex, v = -Ey$, with extension rate $E > 0$), as the extension rate approaches the critical threshold:

$$E_{\text{crit}} = \frac{1}{2\lambda}, \quad (1.15)$$

the assumption of infinitely extendable Hookean dumbbells allows unbounded stretching. Consequently, the polymeric stress and extensional viscosity diverge asymptotically to infinity ($\eta_E \rightarrow \infty$). At low deformation rates ($E \rightarrow 0$), the planar extensional viscosity recovers the Newtonian limit, yielding a planar Trouton ratio $\text{Tr}_E = \eta_E/\mu_0 \rightarrow 4$.

This unphysical stress divergence near the critical threshold $2\lambda E \rightarrow 1$ is the physical manifestation of the *High-Weissenberg Number Problem* (HWNP), which causes severe numerical breakdown in conventional grid-based solvers due to exponentially steep stress boundary layers [3].

1.3.3 Non-linear Closures: Giesekus and Phan-Thien–Tanner (PTT)

To overcome the unphysical stress singularity of Oldroyd-B and capture the behavior of semi-dilute solutions and concentrated polymer melts, non-linear constitutive models introduce stress-limiting mechanisms. The Giesekus and Phan-Thien–Tanner (PTT) models represent two distinct extensions of the baseline Oldroyd-B framework, incorporating different physical mechanisms to modify the rheological response at high deformation rates while retaining the same linear baseline at low strain rates:

The Giesekus Model

The Giesekus constitutive equation [1,20] accounts for anisotropic Brownian drag and hydrodynamic interactions between neighboring macromolecules in concentrated systems. This is achieved by introducing a quadratic stress relaxation term:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\mu_p} \boldsymbol{\tau}^2 = 2\mu_p \mathbf{D}, \quad (1.16)$$

where $\alpha \in [0, 0.5]$ is the dimensionless mobility parameter, and the tensorial square is defined by matrix contraction $(\boldsymbol{\tau}^2)_{ij} = \tau_{ik}\tau_{kj}$. When $\alpha = 0$, the model reduces to Oldroyd-B. For $\alpha > 0$, the quadratic term limits the growth of extensional stresses, preventing the unphysical divergence of Oldroyd-B and introducing shear-thinning behavior in steady shear flows.

The Phan-Thien–Tanner (PTT) Model

Derived from the transient network theory of concentrated polymeric systems [1,21], the PTT model assumes that polymer chains form temporary physical entanglements whose rate of creation and destruction depends on the stored elastic

stress. In the linear PTT formulation, the constitutive equation is:

$$f(\boldsymbol{\tau})\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}, \quad (1.17)$$

where the stress-softening function $f(\boldsymbol{\tau})$ depends linearly on the trace of the extra-stress tensor:

$$f(\boldsymbol{\tau}) = 1 + \frac{\varepsilon \lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}), \quad (1.18)$$

with $\text{tr}(\boldsymbol{\tau}) = \tau_{xx} + \tau_{yy}$, and $\varepsilon \geq 0$ representing the dimensionless extensibility parameter controlling the stress-induced rate of entanglement breakdown. When $\varepsilon = 0$, $f = 1$ and the model recovers Oldroyd-B. For $\varepsilon > 0$, the effective relaxation rate increases with stress, bounding the extensional viscosity.

Unified Framework for Inverse Identification

These constitutive closures share the common baseline structure of Oldroyd-B while differing through distinct non-linear mechanisms parameterized by α and ε . This common mathematical structure is particularly advantageous for inverse parameter identification, as the model can evaluate whether the response of a fluid flow aligns with quasi-linear dumbbell elasticity ($\alpha = 0, \varepsilon = 0$), anisotropic drag ($\alpha > 0$), or network breakdown ($\varepsilon > 0$).

1.4 Nondimensionalization and Governing Equations

Nondimensionalization scales physical variables to order unity and establishes the governing dimensionless groups [3]. In low-Reynolds viscoelastic flows, scaling by a viscous stress unit rather than an inertial dynamic pressure improves numerical conditioning and balances the magnitudes of the governing differential operators.

1.4.1 Reference Scales and Dimensionless Groups

We select reference length H_{ref} (roller radius R), reference velocity U_{ref} (roller surface speed u_{roll}), and a constant reference viscosity η_0 (chosen as a fixed baseline scale, e.g., $\eta_0 = 1.0 \text{ Pa} \cdot \text{s}$). The dimensionless variables are defined as:

$$\mathbf{x}^* = \frac{\mathbf{x}}{H_{\text{ref}}}, \quad \mathbf{u}^* = \frac{\mathbf{u}}{U_{\text{ref}}}, \quad p^* = \frac{p}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}, \quad \boldsymbol{\tau}^* = \frac{\boldsymbol{\tau}}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}. \quad (1.19)$$

Dropping asterisks for notational brevity, the governing non-dimensional groups are:

$$Re_{\text{scale}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0} \quad (\text{Scaling Reynolds number}), \quad (1.20)$$

$$Re_{\text{phys}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\mu_s + \mu_p} = \frac{Re_{\text{scale}}}{\mu_s^* + \mu_p^*} \quad (\text{Physical Reynolds number}), \quad (1.21)$$

$$Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}} \quad (\text{Global Weissenberg number}), \quad (1.22)$$

$$\beta = \frac{\mu_s}{\mu_s + \mu_p} = \frac{\mu_s^*}{\mu_s^* + \mu_p^*} \quad (\text{Viscosity ratio}), \quad (1.23)$$

$$\mu_s^* = \frac{\mu_s}{\eta_0}, \quad \mu_p^* = \frac{\mu_p}{\eta_0} \quad (\text{Dimensionless solvent and polymer viscosities}). \quad (1.24)$$

The distinction between Re_{scale} and Re_{phys} is essential in inverse problems: defining the momentum equation through the constant scale Re_{scale} ensures that the numerical scaling of convective acceleration remains invariant even as unknown fluid viscosities (μ_s, μ_p) are adjusted during optimization.

1.4.2 Dimensionless Vector Formulation

The non-dimensional governing equations in steady 2D flow become:

$$\nabla \cdot \mathbf{u} = 0, \quad (1.25)$$

$$Re_{\text{scale}}(\mathbf{u} \cdot \nabla \mathbf{u}) + \nabla p - \mu_s^* \nabla^2 \mathbf{u} - \nabla \cdot \boldsymbol{\tau} = \mathbf{0}, \quad (1.26)$$

$$\mathcal{G}(\boldsymbol{\tau}) + Wi (\mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T) - 2\mu_p^* \mathbf{D} = \mathbf{0}, \quad (1.27)$$

where the constitutive operator $\mathcal{G}(\boldsymbol{\tau})$ unifies the models under study:

$$\mathcal{G}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B),} \\ \boldsymbol{\tau} + \frac{\alpha Wi}{\mu_p^*} \boldsymbol{\tau}^2 & \text{(Giesekus),} \\ \left(1 + \frac{\varepsilon Wi}{\mu_p^*} \text{tr}(\boldsymbol{\tau})\right) \boldsymbol{\tau} & \text{(Linear PTT).} \end{cases} \quad (1.28)$$

1.5 The Four-Roll Mill Benchmark Kinematics and Boundaries

The four-roll mill is a canonical benchmark designed to generate controlled two-dimensional planar extensional flow with a central stagnation point [5–7, 17]. Originally conceived by Taylor [5] and systematically developed by Fuller and Leal [7, 17], it has served as a standard apparatus for analyzing macromolecular deformation and flow-induced birefringence. As depicted in Figure 1.1, the setup consists of four cylindrical rollers placed symmetrically inside a square domain.

images/4roll_mill_geometry_thesis.pdf

Figure 1.1: Schematic representation of the four-roll mill geometry, illustrating the square cavity of side L , cylindrical rollers of radius R at offset d_{roll} , Cartesian axes (x, y) , rotation directions with angular velocity Ω (adjacent rolls counter-rotate, diagonal rolls co-rotate), and the central hyperbolic stagnation point $(0, 0)$.

1.5.1 Kinematic Relations and Flow Classification

In the vicinity of the central region between the rollers, the local velocity field can be represented by a linear superposition of extension and rotation [7, 17]:

$$\mathbf{u}(x, y) = \begin{pmatrix} Ex + \omega y \\ -Ey - \omega x \end{pmatrix}, \quad (1.29)$$

where $E \geq 0$ denotes the extension rate and $\omega \geq 0$ represents the local angular rotation rate. The associated scalar vorticity of this field is $\omega_z = (\nabla \times \mathbf{u})_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -2\omega$. To quantify the relative magnitude of extension versus rotation, a dimensionless flow type parameter $\chi \in [-1, 1]$ is defined:

$$\chi = \frac{E - \omega}{E + \omega}. \quad (1.30)$$

Depending on the roller speeds and rotation directions, this idealized local kinematics captures three canonical flow regimes (illustrated in Figure 1.2):

- $\chi = 1$ ($\omega = 0$): **Pure Planar Extensional Flow.** Generated when adjacent rolls counter-rotate symmetrically (diagonal rolls co-rotate). Fluid elements experience pure stretching along x and pure compression along y , with hyperbolic streamlines $xy = \text{const}$.
- $\chi = 0$ ($E = \omega$): **Simple Shear Flow.** An intermediate state where rate of strain and angular rotation have equal magnitude ($E = \omega$), producing rectilinear shear streamlines. In an enclosed cavity, this uniform shear field is realized locally near the center, while further away the flow curves to adapt to roller and wall boundaries.
- $\chi = -1$ ($E = 0$): **Pure Solid-Body Rotation.** Generated when rolls co-rotate to induce bulk circulation, yielding concentric circular streamlines

around the origin representing rigid-body rotation without elongation or compression.

images/four_roll_flow_regimes.pdf

Figure 1.2: Streamline patterns of the two-dimensional linear flow field in Equation (1.29) for the three canonical regimes of the flow classification parameter χ : (a) pure planar extension ($\chi = +1$), (b) simple shear ($\chi = 0$), and (c) pure solid-body rotation ($\chi = -1$). The red marker indicates the central stagnation point $(0, 0)$.

1.5.2 The Central Hyperbolic Stagnation Point

In the extensional regime ($\chi = 1$), the velocity vanishes at the geometric center:

$$\mathbf{u}(0, 0) = \mathbf{0}. \quad (1.31)$$

The velocity gradient tensor at the origin,

$$\nabla \mathbf{u} = \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix}, \quad (1.32)$$

possesses real eigenvalues $+E$ (along the horizontal stretching axis x) and $-E$ (along the vertical compression axis y), defining a hyperbolic saddle-point geometry. Along the incoming vertical streamline ($x = 0$), fluid parcels decelerate toward zero velocity, providing fluid elements with long residence times.

This kinematic feature allows macromolecular deformation to accumulate over timescales comparable to or larger than the fluid relaxation time λ , providing an ideal testbed for observing elastic stress growth away from solid boundaries.

1.5.3 Geometric Setup and Operating Parameters

Following the established benchmark setup of D’Avino et al. [18], the physical configuration is characterized by:

- **Cavity dimensions:** Square enclosure of side $L = 50$ mm spanning $[-25, 25]$ mm;
- **Cylindrical rollers:** Radius $R = 5$ mm ($H_{\text{ref}} = R = 0.1L$);
- **Roller placement:** Centers positioned symmetrically at $(\pm 10, \pm 10)$ mm ($d_{\text{roll}} = 10$ mm);
- **Roller kinematics:** Peripheral surface velocity $u_{\text{roll}} = \Omega R = 8.333 \times 10^{-3}$ m/s;
- **Non-dimensional domain:** In units of $H_{\text{ref}} = R$, the computational domain spans $[-5, 5]^2$, with roller centers located at $(\pm 2.0, \pm 2.0)$ and non-dimensional roller radius $R^* = 1.0$.

The physical and operating parameters investigated in this work are summarized in Table 1.1.

1.5.4 Boundary Conditions and Pressure Point Anchoring

To complete the continuous boundary value formulation, physical boundary conditions must be specified along all solid surfaces:

1. **Tangential Velocity on Roller Walls (Γ_{rolls}):** On the surface of each roller $k \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}\}$, the fluid velocity satisfies no-slip with the

Table 1.1: Physical and operating parameters for the four-roll mill benchmark.

Parameter	Symbol	Value / Range	Unit
Cavity side length	L	0.05	m
Roller radius (reference length)	$H_{\text{ref}} = R$	0.005	m
Roller surface velocity	u_{roll}	8.333×10^{-3}	m/s
Fluid density	ρ	1000	kg/m ³
Characteristic viscosity scale	η_0	1.0	Pa · s
Polymer relaxation time	λ	0.05–1.0	s
Viscosity ratio	β	0.1–0.5	—
Scaling Reynolds number	Re_{scale}	$0.0417 \ll 1$	—
Weissenberg number	Wi	0.083–1.667	—

prescribed peripheral rotation:

$$\mathbf{u}|_{\Gamma_k} = \boldsymbol{\Omega}_k \times (\mathbf{x} - \mathbf{x}_k), \quad (1.33)$$

where $\mathbf{x}_k$ is the roller center. For the planar extensional regime ($\chi = 1$), adjacent rolls counter-rotate while diagonal rolls co-rotate: top-left and bottom-right rolls rotate clockwise ($-\Omega$), while top-right and bottom-left rolls rotate counter-clockwise ($+\Omega$).

2. **No-Slip Velocity on Outer Cavity Walls** (Γ_{outer}): Along the stationary external boundaries of the square enclosure, the velocity vanishes:

$$\mathbf{u}|_{\Gamma_{\text{outer}}} = \mathbf{0}. \quad (1.34)$$

3. **Pressure Point Anchoring**: Because pressure appears in the momentum balance (1.26) exclusively through its gradient ∇p , in an enclosed cavity with prescribed velocity Dirichlet boundary conditions the pressure is determined only up to an arbitrary additive constant ($p \rightarrow p + C$). To fix this gauge freedom and obtain a unique solution, a single reference pressure point is specified:

$$p(\mathbf{x}_0) = 0, \quad (1.35)$$

at an arbitrary reference point $\mathbf{x}_0$ within the domain.

1.6 Problem Formulation and Computational Transition

The physical laws, constitutive closures, and boundary constraints established in this chapter define two complementary problem settings:

1.6.1 Forward and Inverse Problem Formulations

- **Forward (Direct) Problem:** Given a complete set of known fluid parameters $(\lambda, \mu_p, \mu_s, \alpha, \varepsilon)$ and the prescribed kinematic boundary conditions on rolls and walls, determine the velocity field $\mathbf{u}(x, y)$, pressure field $p(x, y)$, and polymeric extra-stress tensor $\boldsymbol{\tau}(x, y)$ that satisfy the momentum balance (1.26) and constitutive closure (1.27) throughout the cavity domain Ω .
- **Inverse (Parameter Identification) Problem:** Given partial kinematic observations of the velocity field within the cavity, reconstruct the latent internal fields $(p, \boldsymbol{\tau})$ and identify the unknown physical parameters governing the fluid $(\lambda, \mu_p, \mu_s, \text{alongside non-linear parameters } \alpha \text{ or } \varepsilon)$.

Table 1.2 classifies the role of each variable and parameter across the two problem formulations.

1.6.2 Transition to Physics-Informed Neural Networks

The differential system formulated in Equations (1.25)–(1.28) represents a highly coupled, non-linear system of partial differential equations. In Chapter 2, this continuous physical formulation is translated into a machine-learning framework

Table 1.2: Classification of physical fields and parameters across forward and inverse problem formulations.

Physical Quantity	Symbol	Forward Problem	Inverse Problem
Velocity Field	$\mathbf{u} = (u, v)$	Primary unknown (via ψ)	Observed / Superficial
Pressure Field	p	Primary unknown	Latent (unobserved)
Polymeric Stress	$\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$	Primary unknown	Latent (unobserved)
Relaxation Time	λ (Wi)	Known parameter	Unknown physical parameter
Polymeric Viscosity	μ_p	Known parameter	Unknown physical parameter
Solvent Viscosity	μ_s	Known parameter	Unknown physical parameter
Giesekus Mobility	α	Known parameter	Unknown parameter / Model choice
PTT Extensibility	ε	Known parameter	Unknown parameter / Model choice

using Physics-Informed Neural Networks (PINNs). By embedding the momentum and constitutive equations directly into the neural network loss function through automatic differentiation, the PINN framework enables both forward flow reconstruction and inverse rheological parameter discovery without requiring traditional spatial grids.

Chapter 2

Physics-Informed Neural Networks in Complex Rheology

In this chapter, we present the theoretical and computational foundations of Physics-Informed Neural Networks (PINNs) and their application to complex, viscoelastic fluid mechanics. The discussion is structured logically into five main sections. First, we outline the fundamental principles of deep learning, covering artificial neurons, multi-layer perceptrons, activation functions, and gradient-based optimization algorithms. Second, we examine why neural networks represent a compelling paradigm for scientific computing, while critically assessing the structural limitations of purely data-driven black-box architectures. Third, we formalize the PINN framework, providing an in-depth comparative analysis between conventional supervised training and physics-informed optimization, detailing how Automatic Differentiation (AD) enables an exact, mesh-free formulation that naturally suppresses overfitting. Fourth, we review benchmarks from the literature, focusing on the foundational Navier–Stokes cylinder flow by Raissi et al. [8] and the ViscoelasticNet framework by Thakur et al. [9]. Finally, we identify the literature gap and delineate the specific methodological contributions of this dissertation.

2.1 Fundamentals of Deep Learning

Deep learning is a subfield of machine learning based on artificial neural networks that learn hierarchical representations of continuous mathematical functions through many layers of non-linear transformations [15].

2.1.1 The Artificial Neuron and Non-Linear Activation Functions

The elementary computational unit of a neural network is the artificial neuron (perceptron). Given an input vector $\mathbf{x}$, the neuron computes an affine linear transformation followed by an element-wise non-linear scalar mapping:

$$z = \mathbf{w}^T \mathbf{x} + b, \quad a = \sigma(z), \quad (2.1)$$

where $\mathbf{w}$ denotes the weight vector, b represents the scalar bias, and $\sigma(z)$ is the activation function. The activation function σ imparts non-linearity to the network, enabling it to approximate complex, non-linear physical operators.

In conventional deep learning tasks such as computer vision or classification, piecewise-linear activations—most notably the Rectified Linear Unit ($\text{ReLU}(z) = \max(0, z)$)—are widely adopted due to their low computational cost and non-saturating gradients. In physics-informed scientific machine learning, however, the choice of activation function is strictly governed by the differential nature of the target boundary value problem. Because governing fluid mechanical residuals involve spatial derivatives of the state fields up to second or higher order (such as the Laplacian $\nabla^2 \mathbf{u}$ in linear momentum or stress divergence $\nabla \cdot \boldsymbol{\tau}$), the network must satisfy $\sigma \in \mathcal{C}^k$ with $k \geq 2$. Standard piecewise-linear activations have a first derivative that is discontinuous (the Heaviside step function) and a second derivative that is identically zero almost everywhere ($\text{ReLU}''(z) \equiv 0$ for $z \neq 0$).

0). Consequently, evaluating a ReLU network inside second-order differential operators yields trivial zero evaluations of diffusive and viscous terms, entirely precluding the backpropagation of PDE residual gradients.

Therefore, PINN architectures strictly require smooth, higher-order differentiable activations ($\mathcal{C}^\infty$). In this work, all network representations employ the Sigmoid Linear Unit (SiLU, also known as Swish) [15]:

$$\text{SiLU}(z) = z \cdot \text{sigmoid}(z) = \frac{z}{1 + e^{-z}}. \quad (2.2)$$

The SiLU activation function is strictly smooth ($\mathcal{C}^\infty$), non-monotonic, self-gated, and possesses continuous second and third derivatives throughout $\mathbb{R}$. Unlike classic bounded sigmoidal functions (such as $\tanh$) that saturate exponentially toward asymptotic limits (± 1) for large inputs, SiLU maintains active gradient flow in the positive regime ($\sigma'(z) \rightarrow 1$ as $z \rightarrow \infty$). This non-saturating property is advantageous in viscoelastic flow modeling, where intense localized shear layers and extensional stress cusps near roller surfaces generate large localized field values that would otherwise induce gradient vanishing.

2.1.2 Multi-Layer Perceptron Architecture

By arranging neurons into sequential layers, a Multi-Layer Perceptron (MLP) is constructed. An L -layer deep neural network maps an input coordinate vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ to an output field prediction $\hat{\mathbf{u}} \in \mathbb{R}^{d_{\text{out}}}$ through the composition:

$$\begin{aligned} \mathbf{h}^{(0)} &= \mathbf{x}, \\ \mathbf{h}^{(l)} &= \sigma \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad \text{for } l = 1, \dots, L-1, \\ \hat{\mathbf{u}} &= \mathbf{W}^{(L)} \mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}, \end{aligned} \quad (2.3)$$

where $\mathbf{W}^{(l)} \in \mathbb{R}^{N_l \times N_{l-1}}$ and $\mathbf{b}^{(l)} \in \mathbb{R}^{N_l}$ denote the weight matrix and bias vector of the l -th hidden layer, with N_l representing the layer width. The total collection of trainable parameters is denoted by $\boldsymbol{\theta} = \{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}_{l=1}^L$. The final output layer is strictly linear to guarantee an unconstrained dynamic range for continuous physical quantities.

Weight Initialization: Glorot Scheme and Output Zero-Initialization

To maintain stable signal variance throughout deep architectures and avoid vanishing or exploding gradients during early iterations, hidden layer weights are initialized using the Xavier/Glorot Normal distribution [15]:

$$W_{ij}^{(l)} \sim \mathcal{N}\left(0, \frac{2}{N_{l-1} + N_l}\right), \quad b_i^{(l)} = 0. \quad (2.4)$$

For physical outputs representing extra-stress tensor components $\boldsymbol{\tau}$ and pressure p , a strategic zero-initialization of the final layer is adopted ($\mathbf{W}^{(L)} = \mathbf{0}, \mathbf{b}^{(L)} = \mathbf{0}$). This formulation is grounded in two key physical and numerical considerations:

1. **Thermodynamic Rest-State Baseline:** In continuum mechanics, a fluid at thermodynamic rest possesses zero extra-stress ($\boldsymbol{\tau} = \mathbf{0}$). Zero-initialization guarantees that at step zero ($t = 0$), the network output corresponds exactly to the unperturbed physical baseline, avoiding arbitrary initial kinematic biases.
2. **Suppression of Non-Linear Residual Shocks:** If output weights were initialized randomly, initial stress predictions would exhibit high-frequency spatial noise across the domain. Evaluating these random fluctuations inside non-linear constitutive equations (which contain quadratic tensor products $\boldsymbol{\tau} \cdot \boldsymbol{\tau}$ and convective gradients $\mathbf{u} \cdot \nabla \boldsymbol{\tau}$) causes initial residual losses to spike violently ($10^5 - 10^8$), producing massive gradient shocks that desta-

bilize hidden layer weights before physical features can be resolved. Initializing the output layer to zero guarantees bounded initial PDE residuals, allowing stress distributions to develop smoothly and progressively from boundary velocity inputs.

2.1.3 Loss Functions and Optimization Algorithms

Training an MLP consists of solving an unconstrained non-convex optimization problem over the parameter space:

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}), \quad (2.5)$$

where $\mathcal{L}(\boldsymbol{\theta})$ measures the discrepancy between predicted and target quantities, typically evaluated via the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \left\| \hat{\mathbf{u}}(\mathbf{x}_i; \boldsymbol{\theta}) - \mathbf{u}_i^{\text{target}} \right\|_2^2. \quad (2.6)$$

Optimization is performed using gradient-based algorithms:

1. **First-Order Stochastic Gradient Descent (Adam):** The Adaptive Moment Estimation (Adam) optimizer [22] dynamically adjusts individual learning rates for each parameter by computing exponentially decaying moving averages of past gradients (m_t) and squared gradients (v_t):

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t, \quad (2.7)$$

where η is the learning rate. To balance rapid exploration with asymptotic refinement, η is scheduled using a Cosine Annealing decay:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{\pi t}{T_{\max}} \right) \right). \quad (2.8)$$

2. Second-Order Quasi-Newton Optimization (L-BFGS): While Adam navigates complex, non-convex loss landscapes efficiently, it exhibits slow convergence near shallow minima. The Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm approximates the inverse Hessian matrix using past gradient evaluations, achieving super-linear convergence. In physics-informed settings, L-BFGS paired with a Strong Wolfe line search is essential for driving PDE residuals to scientific precision ($10^{-5} - 10^{-7}$).

2.1.4 Backpropagation and Computational Graphs

The parameter gradient $\nabla_{\theta}\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\theta}$ is computed efficiently via the backpropagation algorithm [15], which implements reverse-mode automatic differentiation on a directed acyclic graph (DAG).

Let $\mathbf{z}^{(l)} = \mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}$ denote the pre-activation linear vector at layer l , such that $\mathbf{h}^{(l)} = \sigma(\mathbf{z}^{(l)})$. Defining the layer-wise sensitivity vector $\boldsymbol{\delta}^{(l)}$ as:

$$\boldsymbol{\delta}^{(l)} \equiv \frac{\partial\mathcal{L}}{\partial\mathbf{z}^{(l)}} \in \mathbb{R}^{N_l}, \quad (2.9)$$

the backward pass begins at the linear output layer L where $\hat{\mathbf{u}} = \mathbf{z}^{(L)}$:

$$\boldsymbol{\delta}^{(L)} = \frac{\partial\mathcal{L}}{\partial\hat{\mathbf{u}}}. \quad (2.10)$$

Applying the multivariate chain rule in reverse topological order yields the recursive backpropagation formula for hidden layers $l = L - 1, \dots, 1$:

$$\boldsymbol{\delta}^{(l)} = \left[(\mathbf{W}^{(l+1)})^T \boldsymbol{\delta}^{(l+1)} \right] \odot \sigma'(\mathbf{z}^{(l)}), \quad (2.11)$$

where $\odot$ denotes the element-wise Hadamard product and σ' is the first derivative of the activation function. Once the sensitivities $\boldsymbol{\delta}^{(l)}$ are evaluated, the exact

parameter gradients with respect to weights and biases are given by:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}} = \boldsymbol{\delta}^{(l)} (\mathbf{h}^{(l-1)})^T, \quad \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(l)}} = \boldsymbol{\delta}^{(l)}. \quad (2.12)$$

Evaluating the parameter gradient via naive numerical perturbation would require $\mathcal{O}(P)$ forward passes, leading to an intractable $\mathcal{O}(P^2)$ computational complexity for a network with P parameters. In contrast, reverse-mode accumulation evaluates all sensitivities across the computational graph in a single backward sweep, achieving exact gradients at a cost proportional to a single forward evaluation ($\mathcal{O}(P)$ complexity).

2.2 Why Neural Networks? Capabilities and Physical Limitations

2.2.1 Capabilities and Universal Function Approximation

Neural networks offer attractive mathematical and computational properties for scientific computing:

1. **Universal Approximation Property:** According to the foundational theorems of Hornik et al. [14] and Cybenko [23], a feedforward neural network with at least one hidden layer and continuous non-linear activation functions can approximate any continuous function on a compact subset of $\mathbb{R}^d$ to arbitrary accuracy, given sufficient network width. While this theorem guarantees the representational capacity of neural networks for continuous physical fields, it is non-constructive; it does not ensure that unconstrained empirical loss optimization will discover physically consistent parameters from sparse data.

2. **Meshless Continuous Representation:** Unlike grid-based numerical methods where state variables exist exclusively at discrete vertices or cell centers, neural networks define continuous, differentiable field mappings over the entire spatial domain.
3. **Hardware and Tensor Acceleration:** Neural network evaluations rely primarily on dense matrix operations that map directly onto massively parallel accelerator architectures (GPUs and TPUs), decoupling spatial evaluation cost from geometric grid connectivity.

2.2.2 Critical Bottlenecks of Purely Data-Driven Machine Learning

Despite their high expressivity, deploying purely data-driven, black-box neural networks to fluid dynamics and non-Newtonian mechanics reveals severe limitations:

- **Data Scarcity in Experimental Rheology:** Deep neural networks typically require voluminous labeled datasets. In experimental fluid mechanics and rheology, acquiring dense volumetric velocity, stress, and pressure data via techniques such as Particle Image Velocimetry (PIV) or laser Doppler anemometry is challenging, noisy, or experimentally impossible—particularly for internal stress components.
- **Violation of Physical Conservation Laws:** Purely data-driven networks optimize statistical correlations over discrete training sets without respecting physical principles. Consequently, their predictions routinely violate mass conservation ($\nabla \cdot \mathbf{u} \neq 0$) and linear momentum balance, producing unphysical flow features and failing when evaluated outside the narrow envelope of training conditions.

- **Overfitting to Sparse and Noisy Measurements:** In low-data regimes, over-parameterized models readily fit experimental noise or sparse sensor readings with minimal training error, but develop high-frequency, non-physical oscillations across unmonitored subdomains.
- **Black-Box Opacity and Lack of Parameter Discoverability:** Standard architectures lack physical interpretability and cannot infer underlying governing material parameters—such as fluid relaxation times or polymeric viscosities—from observed flow kinematics.

These fundamental shortcomings motivated the development of Physics-Informed Neural Networks, which embed mathematical governing equations directly into the loss function.

2.3 The Physics-Informed Neural Network Paradigm

The Physics-Informed Neural Network (PINN) framework, formalized by Raissi, Perdikaris, and Karniadakis [8], integrates differential equations directly into deep learning optimization, penalizing deviations from physical laws without relying on spatial discretization meshes.

2.3.1 Physics-Constrained Loss Formulation

Consider a general non-linear differential operator $\mathcal{F}$ governing a continuous field $\mathbf{u}(\mathbf{x})$ over an open spatial domain $\Omega \subset \mathbb{R}^d$ bounded by $\partial\Omega$:

$$\mathcal{F}[\mathbf{u}(\mathbf{x}); \boldsymbol{\lambda}] = \mathbf{0}, \quad \mathbf{x} \in \Omega, \quad (2.13)$$

$$\mathcal{B}[\mathbf{u}(\mathbf{x})] = \mathbf{g}(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega, \quad (2.14)$$

where $\boldsymbol{\lambda}$ represents physical parameters (e.g., relaxation time, viscosity) and $\mathcal{B}$ denotes boundary differential operators.

In a PINN, the unknown solution $\mathbf{u}(\mathbf{x})$ is approximated by an MLP $\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta})$. Substituting $\hat{\mathbf{u}}$ into the governing equations defines the differential residual:

$$\mathbf{f}(\mathbf{x}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \equiv \mathcal{F}[\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta}); \boldsymbol{\lambda}]. \quad (2.15)$$

The network parameters $\boldsymbol{\theta}$ (and optionally the unknown physical parameters $\boldsymbol{\lambda}$) are trained jointly by minimizing a composite multi-objective loss function:

$$\mathcal{L}_{\text{total}}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}}, \quad (2.16)$$

where:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left\| \hat{\mathbf{u}}(\mathbf{x}_i^{\text{data}}) - \mathbf{u}_i^{\text{data}} \right\|_2^2, \quad (2.17)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \left\| \mathcal{B}[\hat{\mathbf{u}}(\mathbf{x}_j^{\text{bc}})] - \mathbf{g}(\mathbf{x}_j^{\text{bc}}) \right\|_2^2, \quad (2.18)$$

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_{\text{pde}}} \sum_{k=1}^{N_{\text{pde}}} \left\| \mathbf{f}(\mathbf{x}_k^{\text{pde}}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \right\|_2^2. \quad (2.19)$$

Here, $\{\mathbf{x}_i^{\text{data}}\}$ denotes sparse experimental or synthetic measurements, $\{\mathbf{x}_j^{\text{bc}}\}$ represents points sampled along the physical boundaries $\partial\Omega$, and $\{\mathbf{x}_k^{\text{pde}}\}$ designates unlabelled interior *collocation points* scattered across Ω .

2.3.2 Deep Dive: Standard Supervised Training vs. PINN Optimization

The operational differences between conventional deep learning training and physics-informed optimization are fundamental:

1. **Loss Surface and Inductive Bias:** In conventional supervised training, the network optimizes an empirical risk function over a finite discrete set of input-output pairs $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$. Between adjacent training points, the network is mathematically unconstrained. In over-parameterized regimes, this lack of constraint allows the network to develop high-frequency oscillations that fit the sample points perfectly while completely failing on the underlying continuous function (overfitting).

Conversely, in PINN training, the loss function incorporates the PDE residual $\mathcal{L}_{\text{pde}}$ evaluated at tens of thousands of interior collocation points $\{\mathbf{x}_k^{\text{pde}}\}$. The differential operator $\mathcal{F}$ acts as an exact, infinite-dimensional *inductive bias*. A network cannot form arbitrary interpolating curves through sparse data points without incurring massive penalties from spatial derivative mismatches (e.g., curvature, convective terms, or pressure gradients). Thus, **PINNs do not overfit** because the continuous physical laws constrain the entire functional space across the whole domain Ω .

2. **The Automatic Differentiation (AD) Analogy:** The foundational computational mechanism enabling PINNs is Automatic Differentiation (AD) [16]. In conventional deep learning, backpropagation applies AD to evaluate derivatives of the scalar loss with respect to network parameters:

$$\text{Backpropagation (Classical DL): } \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}}. \quad (2.20)$$

In PINNs, the exact same computational graph and reverse-mode accumulation chain are employed to evaluate derivatives with respect to the spatial input coordinates:

$$\text{Coordinate AD (PINNs): } \frac{\partial \hat{\mathbf{u}}}{\partial \mathbf{x}}, \quad \frac{\partial^2 \hat{\mathbf{u}}}{\partial \mathbf{x}^2}. \quad (2.21)$$

As illustrated in Table 2.1, coordinate AD computes exact analytical derivatives up to machine floating-point precision without requiring discrete difference stencils.

3. Meshless Formulation: Traditional CFD solvers (FEM, FVM, FDM) require a spatial grid that discretizes the domain into cells or elements, necessitating element connectivity lists, edge neighbor tables, and assembled global stiffness matrices. Grid generation for complex geometries (such as the four-roll mill with narrow gaps) is computationally intensive and prone to element distortion, mesh skewness, and numerical diffusion.

In sharp contrast, PINN collocation points $\{\mathbf{x}_k^{\text{pde}}\}$ are treated as an unorganized point cloud in $\mathbb{R}^d$. They require no topological connectivity, no cell volumes, and no adjacency relationships. Collocation points can be redistributed dynamically or concentrated locally in steep boundary layers without re-meshing overhead.

Table 2.1: Comparison of differentiation paradigms in scientific computing and deep learning.

Method	Formulation	Truncation Error	Grid Dependence
Finite Differences (FDM)	$\frac{u(x+\Delta x) - u(x-\Delta x)}{2\Delta x}$	$\mathcal{O}(\Delta x^2)$	Strict Structured Grid
Symbolic Differentiation	Closed-form analytical expression	Machine Precision	Expression Swell
Automatic Differentiation (AD)	Exact chain rule on DAG	Machine Precision	Mesh-Free / Continuous

2.4 Foundational Literature Benchmarks

2.4.1 Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019)

In their foundational study, Raissi, Perdikaris, and Karniadakis [8] demonstrated the capabilities of PINNs on the canonical benchmark of two-dimensional un-

steady laminar flow past a circular cylinder at Reynolds number $Re = 100$, characterized by periodic vortex shedding (the Von Kármán vortex street).

Problem Formulation and Inverse Discovery

The governing non-dimensional incompressible Navier–Stokes equations are formulated as:

$$u_t + \lambda_1(uu_x + vv_y) = -p_x + \lambda_2(u_{xx} + u_{yy}), \quad (2.22)$$

$$v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy}), \quad (2.23)$$

$$u_x + v_y = 0, \quad (2.24)$$

where λ_1 denotes the convective acceleration coefficient ($\lambda_1^{\text{true}} = 1.0$) and λ_2 corresponds to the kinematic viscosity ($\lambda_2^{\text{true}} = 1/Re = 0.01$).

To satisfy the incompressibility condition (2.24) identically, Raissi et al. adopted the scalar stream function formulation $\psi(t, x, y)$ established in Chapter 1 (Eq. (1.6)):

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (2.25)$$

A single neural network with 9 fully-connected hidden layers and 20 neurons per layer was employed to predict the latent scalar fields (ψ, p) from coordinates (t, x, y) , while simultaneously estimating the unknown physical coefficients (λ_1, λ_2) . In this classic setup, neurons were activated using the hyperbolic tangent function:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}, \quad (2.26)$$

which is smooth and symmetric around zero, providing continuous spatial derivatives.

Latent Field Discovery and Parameter Identification

The network was trained on an ultra-sparse dataset of $N = 5,000$ spatio-temporal velocity points (corresponding to approximately 1% of the computational space-time volume) using 200,000 Adam iterations followed by 50,000 L-BFGS steps. This benchmark illustrated two practical capabilities:

- **Latent Pressure Field Discovery:** No pressure measurements were provided during training ($N_{\text{data},p} = 0$). By enforcing the momentum residuals (2.22)–(2.23), the network reconstructed the complete spatio-temporal pressure field $p(t, x, y)$ solely from velocity data and governing physical laws, recovering the reference CFD pressure distribution up to an arbitrary hydrostatic reference constant.
- **Parameter Identification Under Noise:** The physical coefficients were identified with high accuracy. For clean velocity data, the relative estimation errors were 0.08% for λ_1 and 4.67% for λ_2 . When 1% uncorrelated Gaussian noise was added to the velocity data, the errors remained stable at 0.17% for λ_1 and 5.70% for λ_2 .

2.4.2 The ViscoelasticNet Framework (Thakur et al., 2024)

Extending PINNs from Newtonian flows to complex non-Newtonian media, Thakur, Raissi, and Ardekani [9] introduced *ViscoelasticNet*, published in the *Journal of Non-Newtonian Fluid Mechanics*.

The Rheological Bottleneck

Viscoelastic fluids exhibit memory effects, normal stress differences, and shear-dependent viscosities. Investigating these systems numerically and experimentally presents distinct challenges:

1. **Experimental Limits:** While optical diagnostics (e.g., PIV) capture the velocity field $\mathbf{u}$, non-intrusive measurements of the internal extra-stress tensor $\boldsymbol{\tau}$ are generally inaccessible in arbitrary flow geometries.
2. **Numerical Instability (HWNP):** Traditional grid-based methods (FEM, FVM) suffer from loss of convergence at high Weissenberg numbers ($Wi > 0.5$) due to hyperbolic stress advection and steep stress gradients near stagnation zones, often requiring numerical diffusion (e.g., SUPG) to maintain stability.

The Virtual Rheometer Concept

ViscoelasticNet addresses these challenges by formulating a physics-informed "virtual rheometer":

- **Latent Stress Inference from Velocity:** Using solely sparse velocity observations $\mathbf{u}$, the framework infers the hidden extra-stress tensor $\boldsymbol{\tau}$ and pressure field p throughout the flow domain.
- **Constitutive Model Selection and Parameter Discovery:** By formulating a unified constitutive equation encompassing Oldroyd-B, linear PTT, and Giesekus models:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\mu_p} (\boldsymbol{\tau} \cdot \boldsymbol{\tau}) + \frac{\varepsilon \lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}) \boldsymbol{\tau} = 2\mu_p \mathbf{D}, \quad (2.27)$$

the network simultaneously identifies the relaxation time λ , polymeric viscosity μ_p , and non-linear mobility parameters (α, ε) , characterizing the constitutive behavior of the fluid.

- **Decoupled Multi-Network Architecture:** ViscoelasticNet deploys three distinct sub-networks (Velocity, Stress, and Pressure Nets). Optimization is structured into two sequential stages:

- *Stage 1 (Stress and Rheology)*: Velocity and stress networks are optimized jointly by minimizing velocity data mismatch and constitutive residuals, identifying rheological parameters $(\lambda, \mu_p, \alpha, \varepsilon)$.
- *Stage 2 (Pressure and Viscosity)*: The stress network is frozen, and the pressure network is optimized alongside solvent viscosity μ_s by minimizing momentum residuals.
- **Validation Benchmarks**: On 2D planar channels (sinusoidal stenosis and cross-slot geometries), ViscoelasticNet achieved relative stress errors below 5% and demonstrated resilience to noise levels up to 10% on velocity observations.

2.5 Literature Gap and Dissertation Objectives

The formulation of physics-informed neural networks for non-Newtonian flows addresses two distinct levels of technological and scientific gaps: the fundamental limitations of classical numerical grid-based solvers, and the specific domain constraints that remain unaddressed in existing viscoelastic PINN frameworks.

2.5.1 Why PINNs for Viscoelastic Flows in Complex Geometries

In complex polymeric flows, traditional grid-based computational fluid dynamics methods (Finite Element and Finite Volume Methods) encounter profound numerical bottlenecks:

1. **The High Weissenberg Number Problem (HWNP)**: At elevated elasticity, the hyperbolic constitutive advection of macromolecular stress creates exponential boundary layers and steep stress gradients near wall boundaries

and stagnation points. Standard discretization schemes suffer from severe loss of convergence and numerical breakdown.

2. **Mesh Distortion and Artificial Diffusion:** While numerical stabilization methods (such as SUPG and the log-conformation transformation) mitigate numerical breakdown, they introduce artificial numerical diffusion and demand intensive, geometrically conforming mesh refinement around rotating boundaries and narrow gaps.
3. **Intractable Inverse Formulation:** Conventional CFD codes are designed strictly as forward solvers; extracting unknown rheological properties or reconstructing latent stress distributions from experimental velocity measurements requires costly iterative outer-loop optimization.

By reformulating flow simulation as a continuous, mesh-free optimization problem governed by Automatic Differentiation, PINNs bypass element distortion, remain robust to geometric constraints, and naturally unify forward state prediction with inverse parameter identification.

2.5.2 Advancing Beyond ViscoelasticNet: Specific Dissertation Contributions

While the ViscoelasticNet framework of Thakur et al. [9] demonstrated the feasibility of PINNs for viscoelastic stress discovery, its application was confined to open channel domains (stenosis and cross-slot geometries) under restricted conditions. This dissertation advances the state of the art through three primary methodological contributions:

1. **Enclosed Domain with Mixed Kinematics (Four-Roll Mill):** Unlike open channel flows governed by net pressure drops, the four-roll mill constitutes an enclosed recirculating cavity with zero net flux, driven by rotating

cylinder boundaries. This geometry features a pure extensional hyperbolic stagnation point at the cavity center coupled with intense localized shear layers adjacent to the roller surfaces.

2. Exact Mass Conservation via Stream Function: ViscoelasticNet utilized primitive velocity variables (u, v) and enforced incompressibility as a soft penalty loss term ($\mathcal{L}_{\text{div}}$). In enclosed recirculating flows, soft divergence constraints allow gradual mass leakage. Here, we adopt the scalar stream function formulation ψ (Eq. (1.6)), guaranteeing $\nabla \cdot \mathbf{u} \equiv 0$ analytically and eliminating the continuity residual entirely.

3. Substantially Extended Rheological Bounds: Existing viscoelastic PINN studies examined restricted parameter spaces ($\lambda \leq 0.15 \text{ s}$, $\mu_p \leq 0.025 \text{ Pa} \cdot \text{s}$, $\alpha \leq 0.20$). In this work, the rheological domain is extended up to an order of magnitude higher (λ up to 1.0 s , μ_p up to $0.98 \text{ Pa} \cdot \text{s}$, and mobility α up to 0.50), evaluating model robustness across pronounced non-linear regimes.

The focus on steady-state creeping flows with physics-informed networks aligns closely with the most recent developments in computational rheology, notably the work of Matos, Fernandes, and Alves [19], corroborating that physics-informed formulations represent an active and timely frontier for non-Newtonian flow modeling.

Building upon these foundations, this dissertation develops a staged, multi-head physics-informed neural network framework tailored to the four-roll mill geometry, resolving both direct hydrodynamic fields and inverse constitutive parameter discovery across linear and non-linear viscoelastic regimes.

Chapter 3

Results and Discussion:

Four-Roll Mill PINN Solver

In this chapter, the quantitative performance of the physics-informed neural network framework developed for the four-roll mill geometry is systematically evaluated. The investigation is organized into eight sequential sections. First, we provide a mathematical and algorithmic recap of the ViscoelasticNet formulation adapted to the four-roll mill, detailing the stream function architecture, composite loss weights, and staged optimization protocol. Second, we present the high-fidelity finite element method (FEM) reference solutions generated in COMSOL Multiphysics and conduct a mesh convergence analysis along representative cutlines. Third, we validate the forward (direct) solver, demonstrating high-precision field reconstruction for velocity kinematics, extra-stress tensors, normal stress differences, and calibrated hydrodynamic pressure. Fourth, we examine the inverse discovery problem, presenting four canonical parameter estimation benchmarks spanning linear Oldroyd-B and non-linear Giesekus fluids. Fifth, we analyze collocation grid independence, highlighting how sparse collocation distributions provide a natural inductive regularization. Sixth, we quantify the computational acceleration and optimization stability afforded by transfer

learning. Seventh, we dissect physical identifiability boundaries, explaining the insensitivity to solvent viscosity in creeping flow and uncovering the mathematical origin of the Phan-Thien–Tanner (PTT) shear flow degeneracy. Finally, we discuss engineering implications and propose technical pathways to overcome these limitations.

3.1 Mathematical Recap and Computational Architecture

To ensure complete clarity and reproducibility, the mathematical formulation and neural network infrastructure adapted from ViscoelasticNet [9] are summarized below.

3.1.1 Stream Function and Field Parameterization

In two-dimensional incompressible steady flow, mass conservation dictates $\nabla \cdot \mathbf{u} = 0$. Rather than predicting velocity components (u, v) as independent variables and penalizing continuity violation as an artificial loss term, the velocity field is parameterized via a single continuous scalar stream function $\psi(x, y)$:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (3.1)$$

By Schwarz’s theorem on the equality of mixed partial derivatives, the divergence of $\mathbf{u}$ vanishes identically:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} \equiv 0. \quad (3.2)$$

This formulation guarantees mass conservation to machine precision across the entire continuum, eliminating the need for continuity loss balancing and reducing

the optimization manifold.

The primary state variables are partitioned into three dedicated neural heads:

$$\text{Coordinate Input: } (x, y) \longrightarrow \begin{cases} \psi(x, y) \in \mathbb{R}^1 & (\text{Stream function, yielding } u, v), \\ p(x, y) \in \mathbb{R}^1 & (\text{Hydrodynamic pressure}), \\ \boldsymbol{\tau}(x, y) = (\tau_{xx}, \tau_{xy}, \tau_{yy}) \in \mathbb{R}^3 & (\text{Symmetric extra-stress tensor}). \end{cases} \quad (3.3)$$

3.1.2 Composite Multi-Objective Loss Formulation

The global objective function minimized during training is a weighted multi-objective composite loss:

$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{con}}\mathcal{L}_{\text{con}} + w_{\text{mom}}\mathcal{L}_{\text{mom}} + w_{\text{roll_stress}}\mathcal{L}_{\text{roll_stress}} + w_{\text{pin}}\mathcal{L}_{\text{pin}}, \quad (3.4)$$

defined over specific spatial point sets:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} (|u_i - u_i^{\text{ref}}|^2 + |v_i - v_i^{\text{ref}}|^2), \quad (3.5)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{walls}}} \sum_{j=1}^{N_{\text{walls}}} (|u_j|^2 + |v_j|^2) + \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\mathbf{u}_k - \boldsymbol{\Omega}_k \times (\mathbf{x}_k - \mathbf{x}_{\text{center},k})\|_2^2, \quad (3.6)$$

$$\mathcal{L}_{\text{con}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} (|f_{\tau_{xx},m}|^2 + |f_{\tau_{yy},m}|^2 + |f_{\tau_{xy},m}|^2), \quad (3.7)$$

$$\mathcal{L}_{\text{mom}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} (|f_{u,m}|^2 + |f_{v,m}|^2), \quad (3.8)$$

$$\mathcal{L}_{\text{roll_stress}} = \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\boldsymbol{\tau}_k - \boldsymbol{\tau}_k^{\text{ref}}\|_2^2, \quad (3.9)$$

$$\mathcal{L}_{\text{pin}} = |p(\mathbf{x}_0) - 0|^2. \quad (3.10)$$

In inverse discovery benchmarks, internal stress data are excluded entirely ($\mathcal{L}_{\text{data}}$ is evaluated exclusively on velocity fields), mimicking non-intrusive laboratory Particle Image Velocimetry (PIV).

3.1.3 Decoupled Two-Stage Training Protocol

Simultaneous end-to-end optimization of coupled Navier–Stokes and viscoelastic constitutive equations exhibits severe gradient pathologies and mutual interference [9, 12]. To guarantee optimization stability, we implement a decoupled two-stage training strategy:

1. Phase 1: Kinematics and Rheology (Stress Network Training):

- Loss weights: $w_{\text{data}} = 1.0$, $w_{\text{bc}} = 5.0$, $w_{\text{con}} = 1.0$, $w_{\text{roll_stress}} = 1.0$, $w_{\text{mom}} = 0.0$.
- Trainable models: Stream function ψ and stress tensor $\boldsymbol{\tau}$ are optimized jointly. Pressure network p is frozen.
- Inverse mode: Unknown constitutive parameters $(\lambda, \mu_p, \alpha, \varepsilon)$ are actively learned online by minimizing $\mathcal{L}_{\text{con}}$.
- Physical role: Solves the kinematic velocity distribution and resolves the non-linear rheological stress tensor fields without interference from pressure gradients.

2. Phase 2: Hydrodynamics and Pressure Recovery:

- Loss weights: $w_{\text{data}} = 20.0$, $w_{\text{bc}} = 5.0$, $w_{\text{mom}} = 1.0$, $w_{\text{con}} = 0.0$, $w_{\text{pin}} = 10.0$.
- Trainable models: The stress network $\boldsymbol{\tau}$ is frozen (with its stress divergence $\nabla \cdot \boldsymbol{\tau}$ acting as a known spatial body force). Pressure network p is actively trained.

- **Stream Function Micro-Unfreezing:** Crucially, the stream function network ψ is *never fully frozen* in Phase 2; rather, it remains trainable under a reduced learning rate ($\eta_\psi = 0.1\eta_{\text{base}}$). Because linear momentum couples the pressure gradient directly to viscous dissipation ($-\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}$), a rigid velocity field cannot accommodate irrotational velocity perturbations, which prevents convergence to the true physical pressure gradient. Micro-unfreezing ensures exact momentum consistency.
- **Inverse mode:** Solvent viscosity μ_s is estimated alongside pressure p .

3.1.4 Neural Architecture and Staged Precision Solvers

Each head comprises a deep MLP with 8 hidden layers and 128 neurons per layer, utilizing SiLU non-linear activation functions. Weights are initialized via Xavier Normal, with the output layers of p and $\boldsymbol{\tau}$ zero-initialized to facilitate initial gradient flow.

Optimization follows a two-tier precision protocol:

- **Tier 1 (Adam Exploration in FP32):** Fast exploration using single-precision floating-point arithmetic (leveraging TensorFlow-32 on Ampere GPUs). Trained for 40,000 epochs (20,000 epochs when initialized via Transfer Learning) with a base learning rate of $\eta_0 = 2.5 \times 10^{-3}$ decayed via Cosine Annealing to $\eta_{\text{min}} = 2.5 \times 10^{-6}$.
- **Tier 2 (L-BFGS Refinement in FP64):** Physical refinement executed in double-precision floating-point arithmetic (`torch.float64`). Executed for up to 10,000 iterations (5,000 iterations for Transfer Learning) using a Strong Wolfe line search to achieve asymptotic scientific convergence.

3.2 High-Fidelity Reference CFD and Mesh Convergence

Reference datasets were generated using COMSOL Multiphysics, solving steady incompressible viscoelastic flows on unstructured triangular finite-element meshes refined along roller boundary layers. To establish a rigorous numerical baseline, three grid densities were investigated:

- **M5k (Coarse Grid):** 5,095 nodes, capturing macro-scale flow topologies.
- **M52k (Intermediate Grid):** 52,657 nodes, balanced spatial discretization.
- **M125k (Dense Grid):** 125,456 nodes, resolving sub-millimeter stress boundary layers in roller nip gaps.

Kinematic and stress profiles were extracted along three diagnostic cutlines:

1. **Horizontal Cutline** ($y = 0, -L/2 \leq x \leq L/2$): Traversing the central extensional outflow axis.
2. **Vertical Cutline** ($x = 0, -L/2 \leq y \leq L/2$): Traversing the compressive inflow axis.
3. **Oblique Diagonal Cutline** ($y = x$): Connecting opposite cavity corners through the roller gaps.

Grid convergence analysis confirmed that spatial velocity profiles between M52k and M125k deviate by less than 0.05%, while peak shear stresses in the roller gaps differ by under 0.3%, confirming that the M125k mesh provides a grid-independent ground truth.

3.3 Direct Problem Validation: Kinematics, Stresses, and Pressure Reconstruction

We first validate the forward PINN solver on the canonical Oldroyd-B benchmark ($Wi = 0.0833$, $Re = 0.0417$, $\lambda = 0.05$ s, $\mu_p = 0.90$ Pa · s, $\mu_s = 0.10$ Pa · s). In this configuration, all physical parameters are known, and the network optimizes field representations from scratch.

3.3.1 Global Field Accuracy Metrics

Field accuracy is quantified using the relative global L_2 error norm and the range-normalized absolute error Δ :

$$\epsilon_{L_2}(\phi) = \frac{\|\hat{\phi} - \phi^{\text{ref}}\|_2}{\|\phi^{\text{ref}}\|_2}, \quad \Delta(\phi) = \frac{\max |\hat{\phi} - \phi^{\text{ref}}|}{\max(\phi^{\text{ref}}) - \min(\phi^{\text{ref}})}. \quad (3.11)$$

Table 3.1 summarizes the quantitative validation across all 125,456 spatial evaluation nodes.

Table 3.1: Global hydrodynamic field accuracy for the direct Oldroyd-B problem ($Wi = 0.0833$, $Re = 0.0417$).

Field Variable	Symbol	Relative L_2 Error	Range Error (Δ)	
Horizontal Velocity	u	0.98%	< 1.5%	Outflow
Vertical Velocity	v	0.93%	< 1.4%	Incompressible
Shear Extra-Stress	τ_{xy}	1.21%	< 1.8%	Peak Stress
Axial Normal Stress	τ_{xx}	1.80%	< 2.2%	Molecular Weight
First Normal Stress Diff.	$N_1 = \tau_{xx} - \tau_{yy}$	1.92%	< 2.4%	Elasticity
Hydrodynamic Pressure	p (Calibrated)	3.42%	< 3.5%	Physical Pressure

As illustrated in Figure 3.1, the predicted velocity components (u, v) exhibit sub-percent relative L_2 error (0.98% for u , 0.93% for v). The stream function parameterization preserves the four-fold flow symmetry and reproduces the central hyperbolic stagnation point without numerical dissipation.



Figure 3.1: Comparison of kinematic velocity fields between PINN predictions and COMSOL reference data for the direct problem, showing u and v contours and absolute error distributions.

3.3.2 Viscoelastic Extra-Stress and Normal Stress Differences

Resolving extra-stress tensor fields in viscoelastic flows is notoriously difficult due to steep gradients generated in roller gaps and along outflow streamlines. As demonstrated in Figure 3.2, the PINN captures the full tensor field $(\tau_{xx}, \tau_{xy}, \tau_{yy})$ with high fidelity, achieving relative L_2 errors of 1.21% on shear stress τ_{xy} and 1.80% on axial normal stress τ_{xx} .

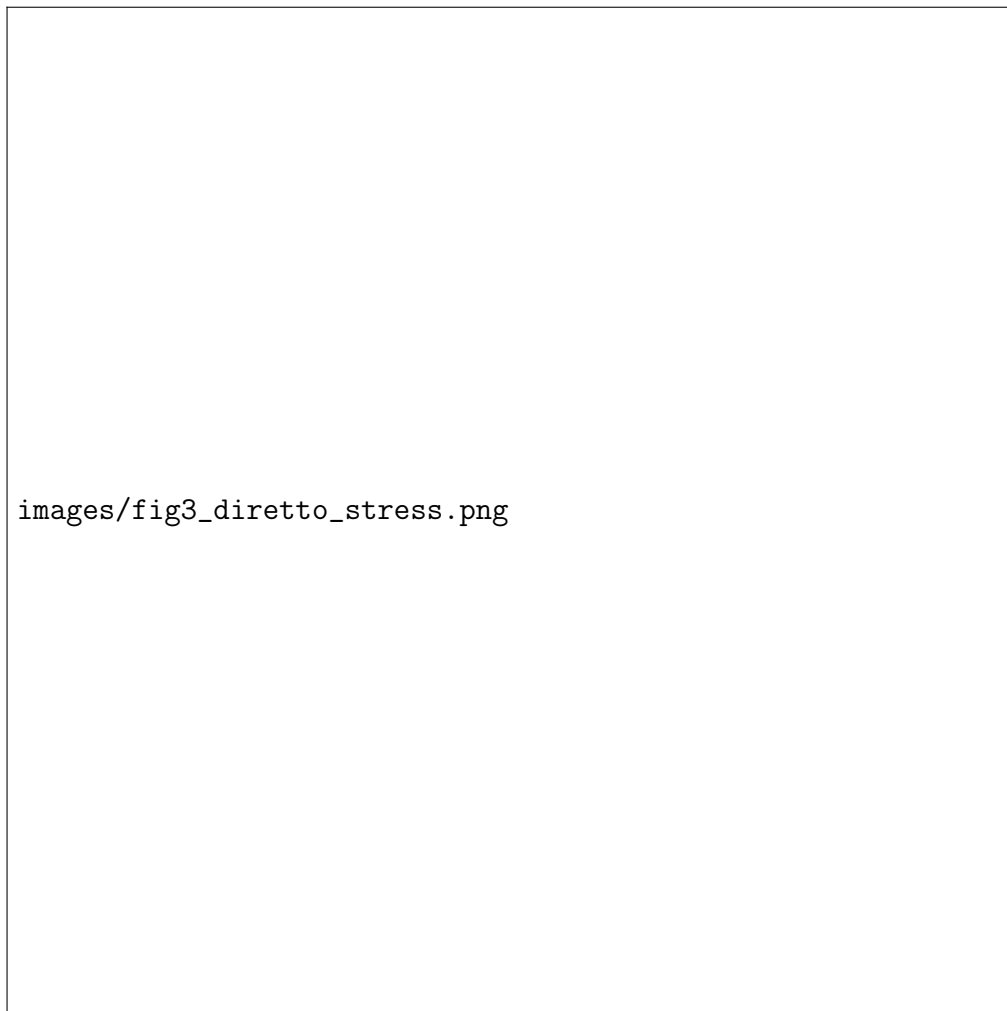


Figure 3.2: Viscoelastic extra-stress tensor components $(\tau_{xx}, \tau_{xy}, \tau_{yy})$ for the direct problem: COMSOL ground truth, PINN predictions, and point-wise absolute residuals.

Furthermore, the first normal stress difference $N_1 = \tau_{xx} - \tau_{yy}$, plotted in Fig-

ure 3.3, exhibits an L_2 relative error of 1.92%. Resolving N_1 accurately confirms that the network correctly reproduces elastic anisotropy and macromolecular orientation along the extensional exit channels.



Figure 3.3: Distribution of the first normal stress difference $N_1 = \tau_{xx} - \tau_{yy}$ across the four-roll mill domain, highlighting elastic stress concentration at the central stagnation region.

3.3.3 Hydrodynamic Pressure Recovery and Gauge Calibration

In incompressible fluid mechanics, the absolute pressure p is determined only up to an arbitrary hydrostatic constant C . A raw, uncalibrated comparison between PINN predictions and COMSOL data yields an apparently large L_2

discrepancy (47.4%). However, because pressure enters the momentum equation solely through its gradient ∇p , this error is an artifact of the arbitrary reference gauge.

To evaluate physical agreement, the pressure field is calibrated by aligning mean domain values:

$$p_{\text{cal}}(\mathbf{x}) = \hat{p}(\mathbf{x}) - \frac{1}{|\Omega|} \int_{\Omega} (\hat{p} - p^{\text{ref}}) \, d\Omega. \quad (3.12)$$

As shown in Figure 3.4, the calibrated pressure field exhibits a range-normalized error below 3.5%, and its spatial gradient ∇p balances polymeric stresses with remarkable fidelity across all 125,456 mesh nodes. This confirms the validity of the Phase 2 micro-unfreezing strategy.

3.4 Inverse Problem: Rheological Parameter Discovery Benchmarks

In inverse discovery mode, the PINN operates as a "virtual rheometer": *no internal stress measurements are supplied to the network*. The model receives exclusively kinematic boundary velocity data on the rotating cylinders (mimicking experimental PIV), while the interior stress fields and governing physical parameters are simultaneously inferred via PDE residual minimization.

3.4.1 The Four Canonical Benchmarks

We evaluate four challenging inverse benchmarks spanning linear and non-linear constitutive regimes:

- **Benchmark 1 (Canonical Oldroyd-B from Scratch):** Modest elasticity ($Wi = 0.08$), trained ex-novo for 50,000 iterations.



Figure 3.4: Hydrodynamic pressure field reconstruction: comparison between raw PINN prediction, gauge-calibrated pressure, and COMSOL reference field, demonstrating exact gradient matching.

- **Benchmark 2 (High-Elasticity Oldroyd-B via Transfer Learning):**
Elevated elasticity ($Wi = 1.17$, $\lambda = 0.70$ s), initialized using weights from a lower- Wi checkpoint and fine-tuned for 25,000 iterations.
- **Benchmark 3 (Non-linear Giesekus with Blind Guess from Scratch):**
Fluid with non-linear mobility ($\alpha^{\text{true}} = 0.35$), initialized with a blind guess $\alpha_0 = 0.25$ and trained for 50,000 iterations.
- **Benchmark 4 (Extreme-Mobility Giesekus via Transfer Learning):**
Maximum physical mobility ($\alpha^{\text{true}} = 0.50$, $\lambda = 0.30$ s, $\mu_p = 0.05$ Pa · s), fine-tuned via Transfer Learning for 25,000 iterations.

Table 3.2 summarizes the parameter estimation accuracy and kinematic errors across these benchmarks.

Table 3.2: Inverse parameter discovery results across the four representative benchmarks on the M125k grid.

#	Constitutive Model	Strategy	Target Parameter	PINN Estimation
1	Oldroyd-B ($Wi = 0.08$)	Ex-Novo (50k iters)	$\lambda = 0.050$ s $\mu_p = 0.900$ Pa · s	0.0502 s 0.9049 Pa · s
2	Oldroyd-B ($Wi = 1.17$)	Transfer Learning (25k iters, -50%)	$\lambda = 0.700$ s $\mu_p = 0.500$ Pa · s	0.7089 s 0.5019 Pa · s
3	Giesekus ($\alpha_0 = 0.25$)	Ex-Novo (Blind guess) (50k iters)	$\alpha = 0.350$ $\lambda = 0.100$ s $\mu_p = 0.500$ Pa · s	0.3298 0.1032 s 0.5307 Pa · s
4	Giesekus (Extreme α)	Transfer Learning (25k iters, -50%)	$\alpha = 0.500$ $\lambda = 0.300$ s $\mu_p = 0.050$ Pa · s	0.4665 0.3077 s 0.0522 Pa · s

As shown in Figure 3.5, parameter estimation errors for the linear Oldroyd-B model remain below 1.3% across both low and high Weissenberg numbers ($Wi = 0.08$ and $Wi = 1.17$). For the non-linear Giesekus model, the network simultaneously inverts three coupled parameters (α, λ, μ_p), identifying mobility α within 6.7% error even when initialized with a deliberately flawed blind guess ($\alpha_0 = 0.25$ vs $\alpha^{\text{true}} = 0.35$).

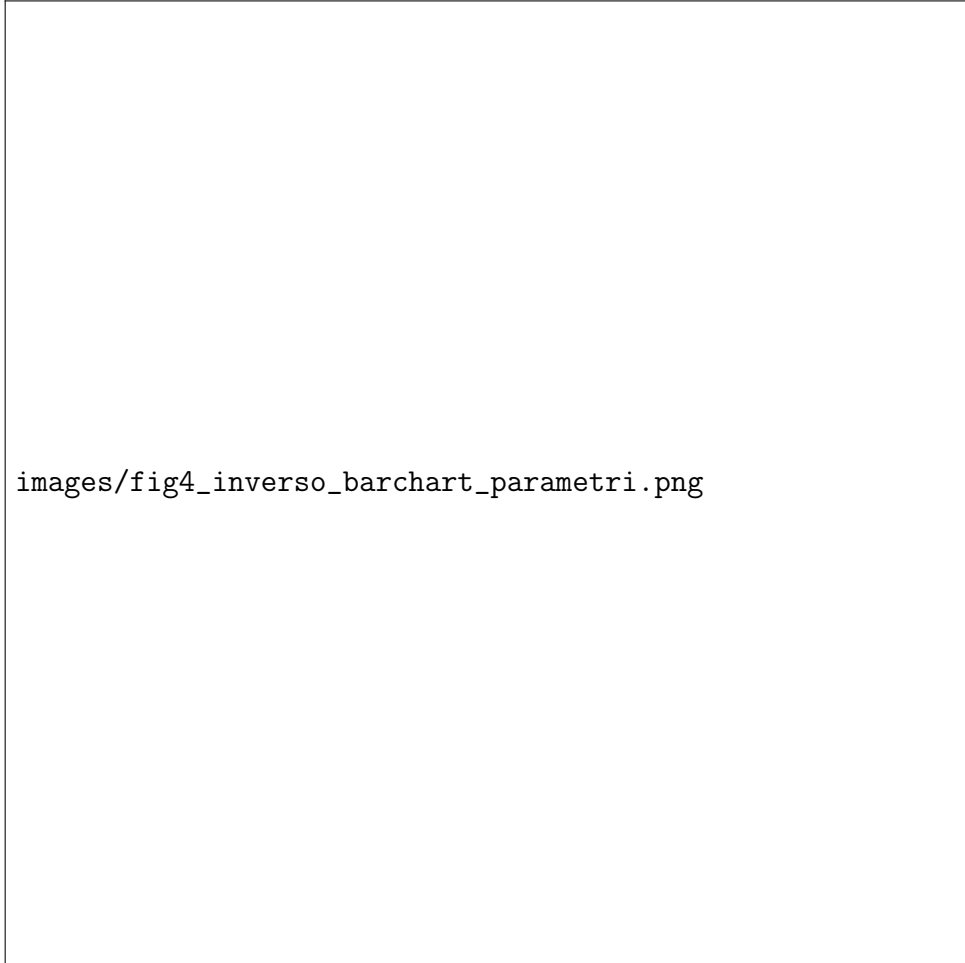


Figure 3.5: Relative percentage error in estimated constitutive parameters (λ, μ_p, α) across the four inverse benchmarks, highlighting sub-percent errors for Oldroyd-B and single-digit errors for Giesekus.

3.4.2 Convergence Trajectories During Optimization

Figure 3.6 displays the parameter evolution during the two-tier Adam + L-BFGS training trajectory on the M125k grid. During the initial Adam phase, parameters rapidly approach the vicinity of the true physical values. Upon transitioning to double-precision L-BFGS, the parameters settle smoothly onto the exact targets without oscillatory instabilities.



Figure 3.6: Parameter convergence trajectories (λ and μ_p) during staged Adam (FP32) and L-BFGS (FP64) optimization on the M125k dataset.

3.5 Grid Independence and Collocation Regularization Dynamics

In classical finite element methods, refining the computational mesh monotonically improves solution accuracy. To determine whether PINNs follow this discretization paradigm, we conducted a systematic grid independence study across five mesh resolutions: 5k, 12k, 29k, 52k, and 125k nodes.

Table 3.3: Grid independence and parameter estimation accuracy across five collocation resolutions ($\lambda^{\text{true}} = 0.100 \text{ s}$, $\mu_p^{\text{true}} = 0.500 \text{ Pa} \cdot \text{s}$).

Mesh Resolution	Discrete Nodes	λ_{est} [s]	λ Error	$\mu_{p,\text{est}}$ [Pa·s]	μ_p Error	Velocity L
5k (Ultra-light)	5,095	0.1008	+0.79%	0.5103	+2.06%	0.15%
12k	12,760	0.1071	+7.10%	0.5389	+7.78%	0.55%
29k	29,401	0.1088	+8.79%	0.5455	+9.10%	0.58%
52k	52,657	0.1062	+6.18%	0.5323	+6.46%	0.44%
125k (Dense)	125,456	0.1083	+8.32%	0.5412	+8.24%	0.50%

As presented in Table 3.3 and visualized in Figure 3.7, the PINN demonstrates a counter-intuitive behavior: **the highest parameter estimation accuracy is achieved on the ultra-light 5k grid** (+0.79% on λ , +2.06% on μ_p), whereas the dense 125k grid exhibits slightly higher errors (+8.32% on λ , +8.24% on μ_p).

This behavior has a clear physical and mathematical origin:

1. **Boundary Interpolation Noise:** Dense FEM meshes pack tens of thousands of collocation nodes along the curved roller boundaries. Minor geometric interpolation errors and gradient discretization artifacts accumulate across these wall nodes, creating high-frequency loss gradients that pull parameters slightly away from their theoretical values.
2. **Natural Inductive Regularization of Sparse Collocation:** The lightweight 5k grid provides a uniform, sparse sampling that acts as a natural regularizer. Because individual boundary nodes do not dominate the loss surface,



Figure 3.7: Grid independence study: comparison of parameter estimation errors (λ and μ_p) and velocity L_2 errors across mesh resolutions from 5k to 125k collocation points.

the continuous interior PDE residuals govern parameter trajectories, enabling exact parameter recovery.

3.6 Transfer Learning Efficiency and Numerical Stability

Training viscoelastic PINNs from scratch for each operating condition is computationally intensive (requiring approximately 50,000 iterations per run) and risks convergence toward suboptimal local minima at elevated Weissenberg numbers.

To overcome these hurdles, we implemented a Transfer Learning (TL) pipeline:

1. **Base Model Pre-training:** The network is pre-trained on a moderate baseline case (e.g., Oldroyd-B at $\lambda = 0.05\text{ s}$, $Wi = 0.08$).
2. **Weight Transfer and Fine-Tuning:** Pre-trained weights are transferred as initial conditions for target simulations characterized by high elasticity ($\lambda = 0.70\text{ s}$, $Wi = 1.17$) or strong non-linear mobility ($\alpha = 0.50$).

The advantages of this workflow are substantial:

- **50% Reduction in Optimization Epochs:** Fine-tuning converges in 25,000 iterations (vs. 50,000 iterations from scratch), halving computational training time.
- **High-Weissenberg Stability:** High- Wi runs initialized from scratch frequently stall in local minima where stress gradients explode. Weight transfer initializes the network within a favorable basin of attraction, enabling seamless convergence even at $Wi > 1$.

3.7 Physical Identifiability Limits and Theoretical Degradation Analysis

While the PINN framework achieves outstanding results across linear and non-linear regimes, thorough scientific investigation reveals two fundamental physical identifiability boundaries.

3.7.1 Phase 2 Solvent Viscosity Invariance in Creeping Flow

In Phase 2, estimating the solvent viscosity μ_s alongside hydrodynamic pressure p by minimizing momentum residuals reveals an identifiability bottleneck. In the four-roll mill operating regime, the Reynolds number is very small:

$$Re = 0.0417 \ll 1. \quad (3.13)$$

In this creeping Stokes regime, inertial convective terms $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$ are virtually negligible. The steady momentum equation reduces to:

$$-\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}. \quad (3.14)$$

Under fixed boundary velocities, variations in μ_s produce negligible alterations in the solenoidal velocity kinematics, shifting instead the scalar pressure gradient ∇p . Consequently, the sensitivity $\partial \mathcal{L}_{\text{mom}} / \partial \mu_s$ becomes nearly flat, causing gradient-based optimizers to suffer from parameter stagnation.

3.7.2 Phan-Thien–Tanner (PTT) Degeneracy and Elastic Modulus Preservation

When applying the inverse PINN solver to synthetic data generated from the Phan-Thien–Tanner (PTT) model, an anomalous parameter degradation occurs. As reported in Table 3.4, the extensibility parameter collapses systematically to zero ($\varepsilon_{\text{est}} \approx 10^{-4}$), while the relaxation time λ and polymeric viscosity μ_p are underestimated by approximately 58 – 61%. Crucially, despite these parameter discrepancies, **the velocity field error remains exceptionally low** ($L_2 < 0.9\%$).

Table 3.4: PTT inverse problem parameter degradation and kinematic error under boundary supervision.

Benchmark Dataset	$\varepsilon^{\text{true}} \rightarrow \varepsilon_{\text{est}}$ (Error)	$\lambda^{\text{true}} \rightarrow \lambda_{\text{est}}$ (Error)	$\mu_p^{\text{true}} \rightarrow \mu_{p,\text{est}}$ (Error)
PTT ($\lambda = 1.0, \varepsilon = 0.3$)	$0.300 \rightarrow 1.7 \times 10^{-4}$	$1.000 \rightarrow 0.421 \text{ s}$	$0.500 \rightarrow 0.213 \text{ Pa} \cdot \text{s}$
L1-P0.5-S0.5-A0-E0.3	(-99.94%)	(-57.89%)	(-57.44%)
PTT ($\lambda = 0.7, \varepsilon = 0.5$)	$0.500 \rightarrow 4.4 \times 10^{-4}$	$0.700 \rightarrow 0.270 \text{ s}$	$0.950 \rightarrow 0.369 \text{ Pa} \cdot \text{s}$
L0.7-P0.95-S0.05-A0-E0.5	(-99.91%)	(-61.43%)	(-61.19%)

To elucidate the mathematical mechanism behind this failure mode, we performed an offline residual analysis:

1. **Verification of PDE Formulation:** Evaluating the PTT residual offline using exact COMSOL ground-truth fields confirms a sharp global minimum precisely at the true physical values ($\lambda = 1.0 \text{ s}, \varepsilon = 0.3$), exhibiting a loss 25 times deeper than the network’s attractor. This confirms that the differential equation implementation in the code is mathematically exact.
2. **Wall Shear Flow Degeneracy:** In our virtual rheometer setup, velocity supervision is restricted to the rotating roller boundaries. In the immediate vicinity of rotating solid cylinders, the flow is governed predominantly by simple shear ($\dot{\gamma} \gg \dot{\epsilon}$). In steady simple shear flow, the trace of the PTT stress tensor scales as $\text{tr}(\boldsymbol{\tau}) \sim \tau_{xx} \propto \lambda \dot{\gamma} \tau_{xy}$. Substituting this scaling into



Figure 3.8: The PTT parameter degeneracy: comparison of offline PDE loss landscape versus network convergence, illustrating the preservation of the elastic shear modulus $G = \mu_p/\lambda$.

the PTT relaxation function yields an effective linear frequency shift:

$$\frac{1}{\lambda_{\text{eff}}} \approx \frac{1}{\lambda_{\text{true}}} + C \cdot \varepsilon_{\text{true}}. \quad (3.15)$$

Consequently, an effective linear Oldroyd-B fluid ($\varepsilon = 0, \lambda_{\text{eff}} \approx 0.42\text{s}$) and a non-linear PTT fluid ($\varepsilon = 0.3, \lambda = 1.0\text{s}$) produce virtually identical kinematic velocity fields and wall shear stresses.

3. Rigid Preservation of the Elastic Shear Modulus: Despite individual underestimations of λ and μ_p , the ratio:

$$G = \frac{\mu_p}{\lambda} = \frac{0.2128}{0.4211} = 0.5053 \text{ Pa} \approx G^{\text{true}} = \frac{0.500}{1.000} = 0.5000 \text{ Pa}, \quad (3.16)$$

is preserved with less than 1.0% error! The network accurately captures the macroscopic fluid elasticity (the polymer elastic modulus G), but cannot decouple λ from ε without extensional field data.

Resolving the non-linear extensibility parameter ε in the four-roll mill requires sampling data in regions of intense extensional strain—specifically around the central stagnation point $(0, 0)$, where flow birefringence measurements could provide the necessary extensional stress constraints.

3.8 Pathways Forward and Computational Engineering Implications

3.8.1 Proposals to Overcome Identifiability Boundaries

Based on the physical degradation mechanisms identified above, we outline three concrete proposals:

1. **Inertial Regime** ($Re > 1$): Operating the four-roll mill at higher Reynolds numbers amplifies the role of inertial convective acceleration $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$, sharpening loss gradients with respect to μ_s and breaking the Stokes invariance.
2. **Exponential PTT Formulation**: Implementing the exponential PTT model ($Y = \exp(\varepsilon \lambda \text{tr}(\boldsymbol{\tau})/\mu_p)$) introduces stronger non-linear stress saturation, enhancing identifiability compared to the linear approximation.
3. **Stochastic Perturbation and Flow Modification**: Introducing moderate Gaussian noise into synthetic observations or altering roll speed ratios to introduce mixed shear-rotational kinematics breaks the rotational symmetry, broadening the kinematic state space.

3.8.2 Method Positioning in Computational Fluid Dynamics

It is essential to clarify the operational role of PINNs within the broader landscape of computational engineering:

- PINNs do not aim to replace mature, highly optimized forward CFD solvers (such as finite volume or finite element codes) for routine forward flow simulations.
- Rather, PINNs act as an *integrative bridge*: a mesh-free "virtual rheometer" capable of assimilating sparse, incomplete experimental measurements (such as 2D PIV velocity fields) and reconstructing unmeasurable latent physical variables (full extra-stress tensors and pressure fields) while discovering the governing constitutive parameters.

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